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BEHAVIORAL AND NEURAL RESPONSES TO TONE ERRORS IN FOREIGN-ACCENTED MANDARIN

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Abstract

Previous ERP research has investigated how foreign accent modulates listener's neural responses to lexical-semantic and morphosyntactic errors. We extended this line of research to consider whether pronunciation errors in Mandarin Chinese are processed differently when made by a foreign-accented speaker. Because Mandarin is a tonal language, both segmental and suprasegmental errors are possible, and tonal errors are known to be common for second language learners. We evaluated behavioral judgments, and the N400 and late positive component (LPC) while native speakers listened to native and foreign-accented sentences containing tone and rhyme pronunciation errors. We observed effects of foreign accent on judgments, suggesting listeners were prone to detect errors in foreign-accented speech more often in sentences with no critical word deviation, but also less likely to reject critical tone errors produced by the foreign-accented speaker. ERP results showed a main effect of accent on LPCs that suggests a difference in degree for sensitivity to foreign-accented compared to native-accented pronunciation errors, rather than a completely different response pattern. We found no effect of accent on N400s, with statistically significant differences between tone and rhyme errors regardless of speaker accent.

Keywords: accented speech, pronunciation, lexical tones, Mandarin, ERPs

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I. Introduction

Understanding how native listeners process foreign-accented speech has important practical applications, and can provide unique insights into basic speech comprehension mechanisms. Using event-related potentials (ERPs) to measure neural responses to errors in spoken sentences, a number of studies have shown that foreign-accented speech can modulate typical responses to lexical and semantic errors (Hanulíková, van Alphen, van Goch, & Weber, 2012; Romero-Rivas, Martin, & Costa, 2015). These effects have been attributed to differences in listeners' expectations about the linguistic properties of foreign-accented speech, or their increased difficulty processing that speech.

We extended this line of research, using ERPs to examine whether native Chinese listeners respond differently to pronunciation errors made by a native or foreign-accented Mandarin speaker. Because Mandarin is a tonal language, both segmental (consonants, vowels) and suprasegmental (tones) pronunciation errors are possible (J.-Y. Chen, 1999; Wan & Jaeger, 1998). Adult English speakers who learn Mandarin as a second language (L2) are known to have difficulty mastering production of tones (e.g., N. F. Chen, Wee, Tong, Ma, & Li, 2016). With this in mind, we manipulated highly constraining sentences, so that sometimes the critical expected word was mispronounced by either a single tone or a single rhyme (vowel and syllable-final /n/ or /ŋ/), resulting in a nonword. While EEG was recorded, Chinese listeners heard sentences spoken by both a native Mandarin speaker and an American L2 speaker of Mandarin, and judged whether they sounded correct.

Our ERP analyses focus on the N400 and the late positive component (LPC). The N400 is a negative-going deflection that peaks around 400 ms after stimulus onset. N400s can be observed in response to each word in a sentence as it unfolds over time, and will reliably grow more negative in response to more unexpected words (the "N400 effect"). In the sentence processing literature the N400 is typically used as an index of difficulty in access or integration of lexical targets (Kutas & Federmeier, 2000; Kutas & Hillyard, 1980, 1984; Lau, Phillips, & Poeppel, 2008). LPCs are positive-going deflections that occur 'late'—following N400s in sentence processing studies—though with large variation in onset and duration. They can be reliably elicited by a wide-range of mismatches in sentence comprehension, including syntactic violations (Osterhout & Holcomb, 1992; Osterhout, Holcomb, & Swinney, 1994), lexical mismatches (e.g., Romero-Rivas et al., 2015; Schirmer et al., 2005), and phonological mismatches (e.g., stress mismatches in Schmidt-Kassow & Kotz, 2009). We will note relevant functional interpretations of LPC in context below, with the caveat that functional accounts of LPC effects are actively debated (see Leckey & Federmeier, 2019).

We compared N400s and LPCs in listeners' responses to pronunciation errors in Mandarin to determine 1) whether the presence of a foreign accent impacts how listeners process pronunciation errors, and 2) whether tone and rhyme errors are processed in the same way.

Previous ERP studies on accented speech processing

A growing number of ERP studies have examined how a speaker's accent can impact the processing of lexical-semantic and syntactic errors, finding notable differences depending on the nature of the error and of the listeners' previous experience with accented speech (Caffarra & Martin, 2018; Grey & van Hell, 2017; Hanulíková et al., 2012; Romero-Rivas et al., 2015; Xu, Abdel Rahman, & Sommer, 2019).

In general, effects of accent tend to be attributed to two main causes (cf. Lev-Ari, 2015). First, listeners may hold different expectations about the linguistic properties of foreign-accented speech. Some syntactic errors or pronunciation patterns that would be odd for a native speaker may be considered normal for an accented speaker (Brunellière & Soto-Faraco, 2013; Caffarra & Martin, 2018; Grey & van Hell, 2017; Hanulíková et al., 2012). Second, the accented speech itself may be less intelligible, and thus more difficult to process, requiring more effort and time on the part of listeners unfamiliar with the novel pronunciation patterns (Goslin et al., 2012; Romero-Rivas et al., 2015).

Hanulíková and colleagues (2012) found that the typical LPCs (also called P600s) that occur when Dutch speakers commit grammatical gender violations were not observed when the same violations were made by a Turkish-accented Dutch speaker. They suggested one way to interpret these results is that they show Dutch listeners have learned from broader experience to expect frequent grammatical gender errors from Turkish-accented speakers. This leads to reduced attempts to repair those errors, as indexed by the late positivities.

The same pattern of LPCs occurring in response to native errors, but not foreign-accented ones has since been found for a variety of languages and linguistic features (Caffarra & Martin, 2018; Romero-Rivas et al., 2015; Xu et al., 2019). While providing general support for the role of listener expectations in foreign-accented speech processing, these studies have introduced additional complexities. For example, Grey and van Hell (2017) found that gender mismatched pronouns in English sentences elicited frontal negativities (the Nref) for native English speech, but not for Chinese-accented English. They also conducted an exploratory subset analysis considering listeners' subjective familiarity with Chinese-accented speakers. Listeners who were able to identify the foreign accent as "Asian" showed early negativities in response to pronoun errors, while those who could not identify the accent showed no response. Caffarra & Martin (2018) manipulated the *typicality* of grammatical errors. Whereas second language speakers of Spanish often make grammatical gender errors, they rarely make number errors. Caffarra and Martin found that both types of errors elicited late positive responses for native accented speech, but only the atypical number violation elicited such responses in foreign-accented speech. Follow-up correlational analysis found that the size of late positive responses for typical errors (but not atypical errors) in foreign-accented Spanish was related to listeners familiarity with that type of accented speech. Speakers who were more familiar with it showed weaker late positive responses than those who reported little experience with English-accented Spanish speakers.

While listener expectations can offer a compelling explanation for ERP results, in some cases, it may be more fitting to attribute observed patterns to the role of processing difficulty. While most studies have used only one or two accented speakers from the same language background, Romero-Rivas and colleagues (2015) presented Spanish listeners with accented speech produced by a variety of differently accented L2 speakers (French, Greek, Italian, Japanese). They found a general increase in N400 responses to (correct) words in foreign-accented speech, relative to native speech, though this effect decreased from the first to second half of the experiment. In the second half of the experiment, they also began presenting semantically misfitting words on some trials (e.g., "Coming to Barcelona we always cross a **piano* in the highway.") and found stronger N400 effects for foreign-accented speech than for native Spanish speech. They interpreted these outcomes as an indication of the relative difficulty listeners had in accessing words in foreign-accented speech.

Among accent studies, however, Romero-Rivas and colleagues' findings represent just one of many different patterns of N400 results. Hanulíková and colleagues (2012) found no N400 differences for lexical-semantic violations between accent conditions (see similar results for 'world knowledge' effects in Foucart, Costa, Morís-Fernández, & Hartsuiker, 2020). Grey and van Hell (2017) found N400 effects only for native-accented English semantic errors, with delayed negativities for the same errors in Chinese-accented English. Xu and colleagues (2019) similarly report finding N400 effects only for native-accented German speech errors ('slips of the tongue'), but no significant N400 differences for such errors in Chinese-accented German. Finally, Goslin and colleagues (2012) found weaker N400 effects for (correct) foreign-accented words compared to native and regionally-accented words. These studies differed in *many* ways (e.g., task, number of talkers, nature of lexical targets, proportion of errors, accent of speakers, and target languages). Given this, varied outcomes should not be surprising, though they do make any generalization across studies rather difficult.

The studies reviewed above have examined responses to various lexical and syntactic errors in foreign-accented speech. Many of these errors were typical of the L2 speaker group under investigation. However, no studies have specifically investigated pronunciation errors—arguably on of the most common error types in L2 speech.

Accented pronunciation and pronunciation errors

An obvious property of foreign-accented pronunciation is that it differs from native pronunciation; but not all differences have the same origins within a speaker, or equal consequences for listeners (for broad discussion, see Derwing & Munro, 2015). This leads us to draw a distinction between *accented* pronunciation and pronunciation *errors*.

We conceptualize foreign accent as a shifted pattern of pronunciation in the dimensional space of phonetic features (e.g., F0, duration), with some or all of the features of the target phonological system realized in novel ways (Kleinschmidt & Jaeger, 2015). This may include both phonemic and sub-phonemic patterns. Importantly, this accented shift is systematic, rather than random; a given sound will consistently be realized in a particular form (or range of forms). For example, a Spanish-accented speaker may systematically produce English /i/ (as in ‘sheep) as closer to the vowel /ɪ/ (as in ‘ship’) (Wade, Jongman, & Sereno, 2007). This may cause some difficulty for listeners, but it is not a random speech error on the part of the speaker. This means that, given enough exposure, a listener could learn and adapt to the patterns of this type of foreign-accented speech.

Unlike accent-shifted pronunciation patterns, pronunciation errors often occur without an obvious pattern, except that of being infelicitous. Furthermore, while they may sometimes appear to be shifted versions of the target sound, they may also include *categorical* errors, where one phoneme is swapped for another (e.g., ‘sheep’ produced sounding like ‘shape’). Superficially, this type of speech error resembles ‘slips of the tongue’, which are typical even in native speech (J.-Y. Chen, 1999; Garnham, Shillcock, Brown, Mill, & Cutler, 1981). However, unlike slips of the tongue, the underlying cause of L2 pronunciation errors is incomplete or uncertain knowledge of the phonological composition of an L2 word. Because there is little pattern to such errors, listeners may not be able to adapt so easily.

Tone errors in foreign-accented Mandarin

The distinction between accented pronunciation and pronunciation errors is useful for understanding tone production in foreign-accented Mandarin. Mandarin has four contrastive lexical tones (Figure 1): Tone 1 (T1) is a high level tone, indicated in Pinyin romanization with a level diacritic (¯) ; Tone 2 (T2) is a rising tone (ˊ); Tone 3 (T3) is a low or low-dipping tone (ˇ), and Tone 4 (T4) is a falling tone (ˋ). While swapping tones can lead to different words (e.g., *mā* “mother” vs. *mǎ* “horse), this is not always the case. The majority of words in Mandarin are disyllabic (Duanmu, 2007), so that tone errors will often result in a nonword. For example, when intending to say *mǎyǐ* (“ant”), with a low tone on the first syllable, if the speaker instead produced *māyǐ*, with a high tone, this would be a nonword.

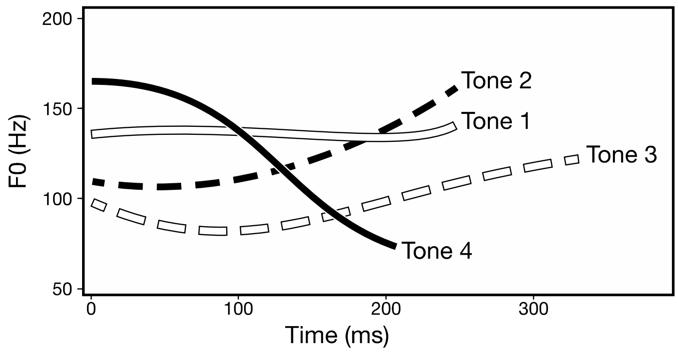

Figure 1. Example of F0 contours for the four citation tones of Mandarin

As already mentioned, native English speakers who learn Mandarin often struggle to master control of lexical tones (Miracle, 1989; Shen, 1989; Sun, 1998; Wang, Jongman, & Sereno, 2003). Just like accents for other phonological features, there may be shifted patterns of tone realizations in an L2 Mandarin speaker's production (Hao, 2018; Wang et al., 2003; Yang, 2016; Zhang, 2016). But when it comes to pronunciation errors, perhaps unlike most segmental errors, multi-directional category substitutions are common for tones in foreign-accented Mandarin, as L2 speakers often swap any one tone for any other (N. F. Chen et al., 2016). This can occur due to interference from native language prosody (White, 1981; Yang, 2016), but can also occur due to misremembered, inaccurately encoded, or completely forgotten tones for specific words (Pelzl, 2018). In other words, in addition to any accented features that may characterize L2 Mandarin speech, frequent pronunciation errors—especially of tones—are an additional characteristic of that speech. If tone errors are frequent enough, a reasonable response from listeners would be to ignore them. However, it is uncertain if listeners can completely ignore pronunciation errors (cf. Pelzl, Carlson, Guo, Jackson, & van Hell, 2020). So then, an alternative possibility is that frequent tone errors will simply make speech comprehension less fluent and more effortful.

Lexical tones in Mandarin speech processing

While there are no previous ERP studies of accented speech in tonal languages, several studies have examined Mandarin and Cantonese listeners' responses to tonal and segmental errors in native speech. Brown-Schmidt & Canseco-Gonzalez (2004) presented listeners with constraining sentences in which they manipulated sentence-final monosyllabic words so that they sometimes mismatched listener expectations. Either segments changed while tones stayed the same (e.g., expected *tāng* 'soup' vs. unexpected *shū* 'book'), tones changed with segments stayed the same (e.g., *tāng* vs. *táng* 'candy'), and both tone and segments changed (e.g., *tāng* vs. *dǒng* 'understand'). The violations elicited increased N400 responses. Brown-Schmidt & Canseco-Gonzalez (2004) found N400 effects for segmental, tonal, and combined violations, with no differences among them. Schirmer and colleagues (2005) obtained the same pattern of results with a similar design using Cantonese sentences and listeners. These studies suggest that native Chinese listeners are equally sensitive to all types of phonological violations, critically, including tones (for similar results with isolated words see Malins & Joanisse, 2012; Zhao, Guo, Zhou, & Shu, 2011).

These studies used monosyllabic critical words and tested responses to vowel or tone changes that produced other meaningful words or morphemes. In other words, the critical violations were simultaneously phonological *and* lexical in nature. Pelzl et al. (2019) also examined tone manipulations but used disyllabic critical words. Pronunciation errors—particularly tone errors—often function differently in disyllabic words because it less likely the error will result in another existing word, whereas monosyllabic errors often do. Pelzl et al. (2019) focused on N400 and LPC responses to tone and rhyme mismatches in L1 and L2 Mandarin listeners. As with the other Chinese ERP studies, Pelzl et al. (2019) also found clear

N400 responses for tone and rhyme mismatches in native Chinese listeners (but not in L2 listeners). However, there were also apparent differences between word and nonword responses. Compared to the expected word condition, the N400 for completely mismatching words remained more negative well beyond the 300-500 ms N400 window they tested, whereas the tone and rhyme responses became strongly positive in the LPC window (450-800 ms), differing significantly from both the expected word and the semantic mismatch condition, suggesting that nonword violations elicited different late processes than real word violations. This contrasts somewhat with Schirmer et al. (2005) who also found strong LPCs after tonal, segmental, or complete (both tone and segmental) violations, but with no differences between violation types. As already suggested, one reason for the differences between studies may be that the violations used by Schirmer and colleagues all resulted in unexpected real words, while those in Pelzl et al. (2019) resulted in nonwords.

To summarize, previous ERP studies suggest that Chinese listeners process segmental and tonal cues in largely the same manner during word recognition, but most research has used violations that were both phonological and lexical. The present study utilizes the design and much of the stimuli of Pelzl et al. (2019) to investigate how rhyme and tone pronunciation errors—resulting in nonwords—are processed in foreign-accented Mandarin.

The present study

The present study considers the potential roles of listener expectations and processing difficulties in the case of foreign-accented Mandarin. We aimed to answer two questions. First, (1) *do native Mandarin listeners process pronunciation errors differently for native and foreign-*

accented speech? We hypothesized that native Mandarin listeners might expect more pronunciation errors of all kinds in foreign-accented speech. This might manifest as smaller LPCs—and perhaps also smaller N400s—for both rhyme and tone pronunciation errors in foreign accented speech. Alternatively, if accented speech increases difficulty in retrieving words, N400s might increase for foreign-accented pronunciation errors compared to the same errors in native speech. Our second question was: (2) *Do native Mandarin listeners process pronunciation errors differently for tones and rhymes*? Given the particular difficulty L2 speakers have with lexical tone, we hypothesized that listeners would have heightened expectations for tone errors compared to rhyme errors. This might manifest as a greater reduction in N400 and/or LPC amplitude for tone errors compared to rhyme errors in foreign-accented speech.

To answer these questions, we conducted an ERP study with native Mandarin Chinese listeners in Beijing, China. Listeners heard sentences produced by either a native speaker of Mandarin (native accent), or an L2 speaker (foreign accent). Sentences were highly constraining and pushed listeners to expect a specific critical word. We compared ERPs to critical words that were expected with mismatching words that had a rhyme mismatch, a tone mismatch, or a complete word mismatch.

Participants

Forty-two native speakers of Mandarin Chinese were recruited from Beijing Normal University and surrounding communities. All participants were from Mandarin speaking regions of China, were right-handed, and had no reported history of language or hearing impairment. Additionally, they were highly educated (either currently in college or already graduated). After data processing, eight participants were removed from later analysis due to excessive EEG artifacts, leaving a final sample of 34 participants (17 male, 17 female; mean age: 23, sd = 2.7). All participants gave informed consent and were compensated for their time. All procedures were approved by the University of Maryland's IRB.

ERP experiment: Design and materials

To investigate the research questions, an auditory ERP sentence experiment was designed with a behavioral sentence judgment task accompanying sentence trials.

Auditory materials consisted of 240 constraining sentences, each occurring with four critical word conditions (expected word, tone mismatch, rhyme mismatch, and word mismatch) and two accent conditions (native, foreign). Figure 2 illustrates the experimental paradigm. The expected word condition (*kèrén* 'guest') provides a reduced N400 baseline for highly predictable words. The word mismatch condition (*zázhì* 'magazine') should induce the typical increased N400 effect relative to expected words. The tone mismatch condition tests N400 effects for mispronounced tones (e.g., a rising tone on *k**é**rén* vs. the expected falling tone on *kèrén*). The rhyme mismatch tests N400 effects for mispronounced rhymes (e.g., *k**ù**rén* vs. the expected kèrén). The rhyme manipulation always affected vowels, but in cases also affected word-final consonants (see Appendix S1 for more details; Appendix S5 for full list of stimuli). Because these last two conditions use nonwords, N400 responses may differ from those in the word mismatch condition (regardless of accent condition) due to the lack of lexical meaning and the correspondence of the second syllable with that of the expected word, which may ease recovery of that word.

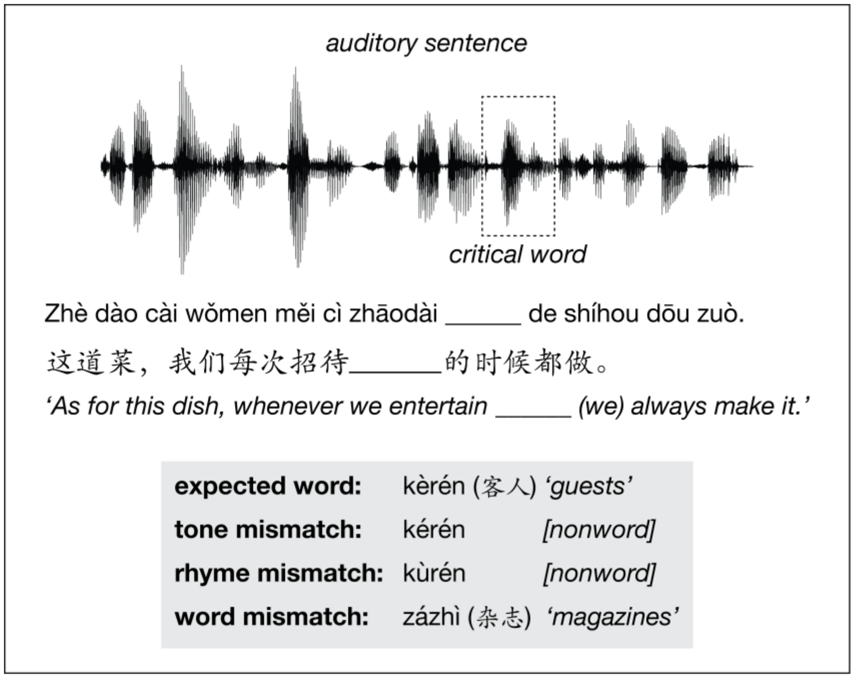

Figure 2. Example of stimulus sentence and critical words (full list of stimuli available in supplementary materials)

Each participant heard 60 sentences in each critical word condition, half produced by a native Mandarin speaker from Beijing, and half produced by an advanced L2 speaker from the United States.[1]

ERP experiment: Procedures

The experiment was conducted in a quiet room in the EEG lab at Beijing Normal University (BNU). Participants were seated in front of a computer with two audio speakers (Edifier R1600TIII) placed to the left and right of the display computer monitor. The experiment was run using Matlab (Mathworks, USA, 2013) and the Psychophysics Toolbox extensions (Brainard, 1997; Pelli, 1997). Instructions were delivered on the computer monitor in Chinese and comprehension was verified orally in Chinese. After participants were fitted with the EEG cap, they were shown the output of the EEG on the monitor and familiarized with the impacts of blinks and other movements on the EEG signal.

Prior to beginning the experiment, participants were familiarized with the accompanying behavioral sentence judgment task. The experimenter explained the three types of errors that might occur in sentences. Participants then completed eight practice sentences not used in the experiment itself, half spoken by a native speaker, half spoken by a non-native speaker. After each sentence, participants were required to answer the question: *Zhege juzi tingqilai shifou zhengque?* ('Did this sentence sound correct or not?'). Participants answered by pressing 'J' for yes and 'F' for no. To illustrate what was intended by the Chinese word *zhengque* ('correct'), feedback was provided after each practice sentence, both in written form presented on the monitor, as well as by oral feedback from a native speaking Chinese lab assistant. Feedback was intended to make clear that only three types of error should be construed as 'not correct', namely, a semantically misfitting word (*bu he yujing* "does not fit context"), a word with its rhyme mispronounced (*fayin bu dui* "pronunciation is incorrect"), or a word with its tone mispronounced (*yudiao/shengdiao bu dui* "tone is incorrect"). Feedback drew attention to the specific word on which an error occurred and what type of error had occurred.

Timing parameters are illustrated in Figure 3. Participants were asked to delay responses until the prompt, 2000 ms after the end of the auditory stimulus. After every 15 trials, participants were given a self-paced break. The entire experiment included 240 sentences and lasted approximately 1 hour, not including preparation of the electrode cap.

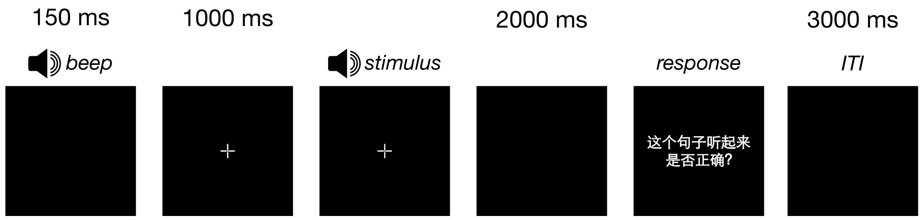

Figure 3. Trial structure and timing parameters. ITI stands for inter-trial interval.

Data processing

Raw EEG was recorded continuously at a 1000 Hz sampling rate using a Neuroscan SynAmps data acquisition system and an electrode cap mounted with 29 AgCl electrodes at the following sites: midline: Fz, FCz, Cz, CPz, Pz, Oz; lateral: FP1, F3/4, F7/8, FC3/4, FT7/8, C3/4, T7/8, CP3/4, TP7/8, P4/5, P7/8, and O1/2. Recordings were referenced online to the right mastoid and re-referenced offline to averaged left and right mastoids. The electro-oculogram (EOG) was recorded at four electrode sites: vertical EOG was recorded from electrodes placed above and below the left eye; horizontal EOG was recorded from electrodes situated at the outer canthus of each eye. Electrode impedances were kept below 5kΩ. The EEG and EOG recordings were amplified and digitized online at 1 kHz with a bandpass filter of 0.1-100 Hz.

For each trial for each participant, we extracted epochs time-locked to the onset of the critical word from -100:1000ms, using preprocessing routines from the EEGLAB v 10.2.5.8b (Delorme & Makeig, 2004) and ERPLAB v3.0.2.1 (Lopez-Calderon & Luck, 2014) toolboxes to reject trials with ocular or muscular artifacts. Muscle potential, sweat, and alpha wave artifacts were identified using the peak-to-peak artifact rejection routine (a 200-ms window was moved across the data in 100-ms increments and any epoch where the peak-to-peak voltage exceeded 100 μV was rejected), and the step function artifact rejection routine was used to identify eye-blink (40 μV threshold) and eye-movement (25 μV threshold) artifacts, followed by visual confirmation of the identified artifacts by the experimenters. Eight participants were removed from further analysis due to having greater than 40% of their trials rejected due to artifacts. Trial-level data was exported to R (v. 3.6.1, R Core Team, 2019) for further analysis. Prior to model fitting, extreme amplitude values above |50| were excluded. For the remaining data, in the N400 window 62,882 (86%) of 73,440 trials were retained (*Total trials: Native accent* expected word = 7953; tone mismatch = 7829; rhyme mismatch = 7813; word mismatch = 7868; *Foreign accent* expected word = 7928; tone mismatch = 7789; rhyme mismatch = 7772; word mismatch = 7930); in the LPC window 62,356 (85%) of 73,440 trials were retained (*Total trials: Native accent:* expected word = 7862; tone mismatch = 7760; rhyme mismatch = 7733; word mismatch = 7815; *Foreign accent:* expected word = 7877; tone mismatch = 7728; rhyme mismatch = 7697; word mismatch = 7884).

Behavioral sentence judgment task

Analysis

We fit mixed-effects logistic regression models with crossed random effects for subjects and items using the *mixed()* function from the *afex* package (v. 0.25-1, Singmann et al., 2017), and the *lme4* package (v. 1.1-21, Bates et al., 2015) in R, with the setting "family=binomial". The dependent variable was response type (binary: 'yes'=1, 'no'=0). Fixed effects included critical word (expected word, tone mismatch, rhyme mismatch, word mismatch), speaker accent (native, foreign), and their interaction. We assume that people will vary in both their responses to mismatch types, their responses to accented speech, and the way these factors interact. As our critical words occurred in sentences, we also expected by-item (sentence) variability as responses to sentences would vary by critical word and accent. Based on this reasoning, we aimed to model by-subject and by-item random intercepts, and by-subject random slopes for the interaction of word and accent, and by-item random slopes for the effect of word and accent. Effects coding was applied using the *mixed()* function in *afex*, and p-values were obtained using the likelihood ratio test ("LRT") method. Model results are reported below as Chi-square tests from a mixed-effects ANOVA, which allows for a convenient summary of main effects and interactions in mixed-effects models. Models were run with the *bobyqa* optimizer in *lme4* (Bates et al., 2015). Our goal was to retain the best fitting model with the maximal random effects structure described above (Barr, Levy, Scheepers, & Tily, 2013). The maximal model failed to converge, so we ran models with suppressed correlations for random effects, and settled for a slightly less than maximal model (*lmer* formula: *score ~ word * accent + (word + accent* || *subj) + (word * accent* || *item)*. In order to examine possible adaptation effects, we also tested models including factors for experiment trial and experiment half, however, these more complex models failed to improve model fit as indicated by likelihood ratio tests, so we will not consider them further here.

Results

The average percent of sentences judged acceptable ('yes' responses) in the sentence judgment task is shown in Table 1. Descriptively, participants tended to accept expected word sentences from the native accented speaker, while rejecting sentences that contained mismatch words. The trends are generally similar for the sentences from the foreign-accented speaker, except that listeners also often rejected his sentences when there was no mismatch manipulation (i.e., for expected word sentences), suggesting they were may have been sensitive to other aspects of the sentence besides the critical word. There was large variation across participants, especially for foreign-accented speech.

Table 1. Percentage 'sentences judged acceptable ('yes' response) for the sentence judgment task. Standard deviations in parentheses.

	expected word	tone mismatch	rhyme mismatch	word mismatch
native accent	91 (29)	14 (35)	7 (26)	5 (22)
foreign accent	58 (49)	24 (42)	11 (32)	3 (17)

Model estimates of sentence acceptance rates are depicted with boxplots and violin plots in Figure 4. Results were submitted to a mixed model ANOVA which indicated a statistically significant main effect of word, $\chi_2(3)=199.20$, $p<.001$, and of accent, $\chi_2(1)=4.30$, $p=.038$, and a significant word-by-accent interaction, $\chi_2(3)=202.30$, $p<.001$.

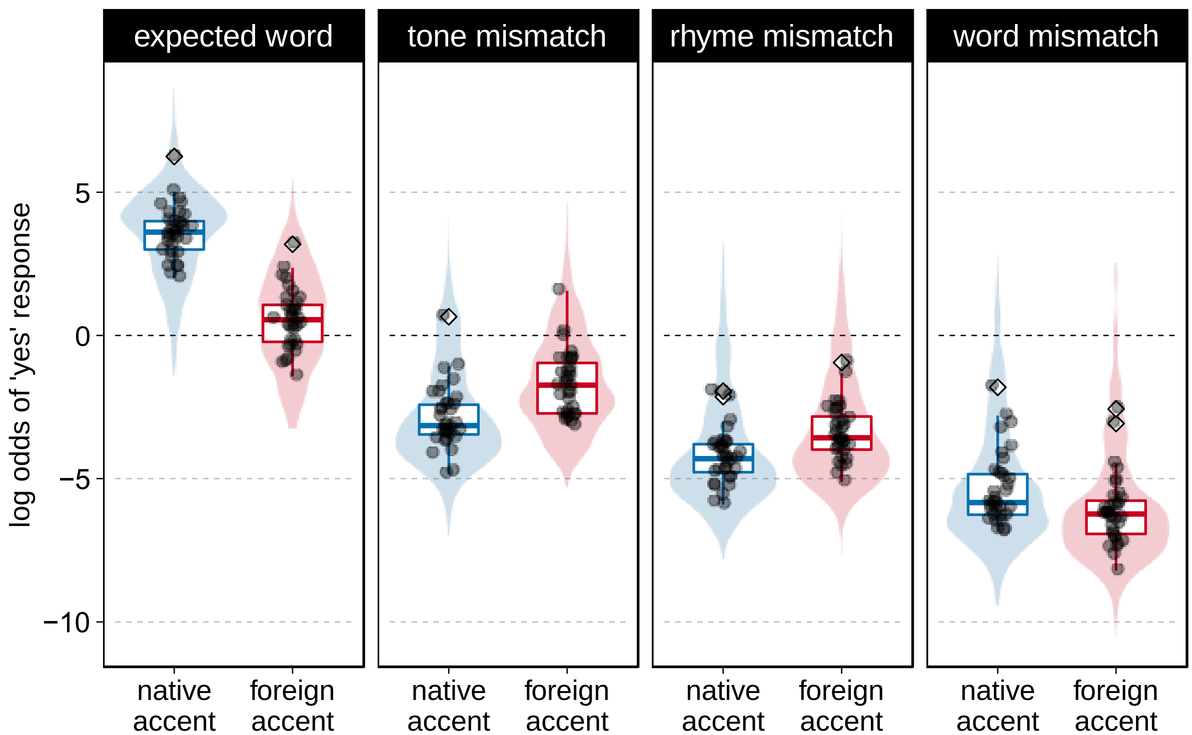

Figure 4. Model estimated log odds of accepting a sentence ('yes' response) for the sentence judgment task. Shaded areas behind the boxplots indicate the distribution of responses. Grey dots indicate the mean of a single participant's responses; white diamonds indicate that a participant's mean is an outlier.

To address our research questions, we conducted post-hoc comparisons using the *multcomp* package (Hothorn, Bretz, & Westfall, 2008), with the Bonferroni-Holm procedure used to correct for multiple comparisons (Table 2). To test whether accent affected listeners' processing of pronunciation errors, we compared the difference in the size of model estimates for judgments of critical word conditions between accents. We will call this difference the 'accent effect'.

Table 2. Post-hoc comparisons for behavioral sentence judgment task model

					Conf. Intervals	
	b	*SE*	*z*	*Pr(>|z|)*	Lower	Upper
Accent effect						
Native: Exp – Foreign: Exp	3.18	0.30	10.75	<.001	2.36	4.00
Native: Tone – Foreign: Tone	-1.31	0.29	-4.51	<.001	-2.11	-0.50
Native: Rhyme – Foreign: Rhyme	-0.79	0.34	-2.34	.058	-1.72	0.15
Native: Word – Foreign: Word	0.72	0.46	1.56	.239	-0.56	2.00
Interactions of accent effects						
(Nat: Exp-Word) – (For: Exp-Word)	2.46	0.53	4.68	<.001	1.00	3.92
(Nat: Tone-Word) – (For: Tone-Word)	-2.03	0.52	-3.89	<.001	-3.47	-0.58
(Nat: Rhyme-Word) – (For: Rhyme-Word)	-1.50	0.55	-2.74	.025	-3.03	0.02
(Nat: Tone-Rhyme) – (For: Tone-Rhyme)	0.52	0.36	1.43	.239	-0.49	1.53
Tone and rhyme effects across accents						
Tone – Rhyme	-3.41	0.62	-5.51	<.001	-5.13	-1.70

p-values adjusted by Bonferroni-Holm method; asymptotic confidence intervals reported
Nat = Native accent; For = Foreign accent;
Exp = expected word; Tone = tone mismatch; Rhyme = rhyme mismatch; Word = word mismatch

We found a large and statistically significant accent effect for the expected word condition, with listeners less likely to accept sentences produced by the L2 speaker. This result may seem surprising, but is reasonable in the context of the task. Though our experimental manipulation was tied to the critical words, listener responses were based on entire sentences. Thus, a listener could reasonably reject expected word sentences due to perceived infelicities in pronunciation *anywhere in the sentence*. In this light, results suggest some listeners had difficulty accepting any sentences produced with a foreign accent. We will return to this issue in the general discussion.

To answer our first research question, whether Mandarin listeners respond differently to pronunciation errors in different accents, we examined the accent effect for mismatch conditions. We found a statistically significant accent effect for sentences containing tone mismatches, with listeners more likely to accept such sentences when produced by the foreign-accented speaker.[2] At the same time, we failed to find significant accent effects for sentences with word or rhyme mismatches—though rhymes trended towards a difference.

To better understand the relationships between word and accent conditions, we examined the interactions between accent effects and conditions (e.g., difference in size of accent effects for rhymes vs. tones). Typically, we would use the expected word condition as a baseline for these comparisons, but because of the large accent effect for expected word sentences, this would be comparing apples and oranges. In contrast, the accent effect for word mismatch sentences was very small, making it a suitable choice as a baseline for testing interactions. As shown in Table 2, the interaction effect indicated that the accent effect for expected word sentences was different than the effect for word mismatches, such that foreign-accented sentences with expected words were more likely to be rejected. The interaction effect for tone and word mismatches also differed, such that foreign-accented sentences with tone mismatches were more likely to be judged acceptable. The interaction effect for rhyme and word mismatches also differed, again indicating that rhyme mismatches were more likely to be judged acceptable in the foreign accent. To summarize, these results further support a tendency for listeners to judge foreign-accented speech as infelicitous more often than native speech in sentences with expected words, but also to be more accepting (or less sensitive) when judging sentences with foreign-accented tone and rhyme mismatches.

To answer our second research question, about possible differences in accent effects for tone and rhyme mismatches, we conducted two additional comparisons. First, we compared the interaction of accent effects for rhyme and tone mismatches. This interaction failed to reach statistical significance. In order to understand the underlying trend, we further compared tone and rhyme mismatch effects *across* accents. There was a significant difference in log odds, with listeners more likely to accept sentences containing tone mismatches than those with rhyme mismatches. This suggests that, behaviorally, listener judgments are less sensitive for tones compared to rhymes—*regardless of speaker accent*.

Given the nature of the sentence judgment task, these results should be treated with some caution, as listener responses cannot be unambiguously tied to the critical word manipulations. Additionally, results are highly variable across participants. When we look at each individual's average raw accuracy in the expected word condition, it ranges from accepting as few as 21% of accented sentences up to as many as 95%. We will address this variability in more detail in the general discussion.

ERP analysis

To analyze results from the ERP experiment, we fit mixed-effects linear regression models with crossed random effects for subjects and items using the *afex* and *lme4* packages in *R*. Separate analyses were run for the 300-500 ms (N400) and 600-900 ms (LPC) windows over nine posterior electrodes. ERPs were time-locked to the onset of critical words. Analyses included all critical trials, regardless of the accuracy of behavioral judgments, as judgments occurred after the end of a trial and were not necessarily tied to the critical words themselves. The dependent variable was mean amplitude, calculated as a mean for each subject and each item at each electrode averaged across the 300-500 ms and 600-900 ms windows. These windows were selected to capture typical N400 and LPC effects, and follow windows used in Pelzl et al. (2019), though we used longer epochs and thus a slightly adjusted LPC window in the present study (600-900 vs. 450-800) (full model output is available in Appendix S3; for interested readers, a parallel analysis using RM ANOVA is available in Appendix S4).

The spatial dimension of ERP data introduces challenges for mixed-effects models (see discussion in Winsler et al., 2018). To constrain variability across electrodes, as seen in other recent ERP studies that used mixed-effects models, we restricted our analysis to a smaller set of electrodes where N400 and LPC effects are typically observed (for a similar approach, see Kuperberg et al., 2020). We chose nine central posterior electrodes: CP3, CPz, CP4, P3, Pz, P4, O1, Oz, O2.[2]

For both analyses (N400, LPC), fixed effects included critical word (effect coded: expected word, tone mismatch, rhyme mismatch, word mismatch), accent (effect coded: native or foreign), and their interaction. We fit the maximal random effects structure including by-subject and by-item random intercepts and slopes for all fixed effects and their interaction. Results below are reported from *F*-tests for mixed model ANOVAs computed on linear mixed-effects models using the *mixed*() function from *afex*. The *lmer/mixed* formula for the final models was: *mean.amplitude ~ word * accent + (word * accent | subj) + (word * accent | item)*. Possible effects of adaptation were evaluated by also testing models with the additional factors trial (1-240) and half (first, second), but as these failed to improve model fit, we did not pursue them further.

N400

Grand average waveforms averaged across all nine posterior electrodes are shown in Figure 5. Mean amplitudes for word and accent conditions in the N400 window are shown in Table 3. Mean results suggest small differences between accent conditions, and more negative responses for rhyme and word mismatches relative to the expected word condition. Mixed Model ANOVA results showed a significant main effect for word $F(3, 93.22)=11.26$, $p<.001$. The main effect of accent $F(1, 103.78)=0.07$, $p=.800$, and the word-by-accent interaction failed to reach significance, $F(3, 81.13)=0.29$, $p=.831$. Model estimates are depicted in Figure 6.

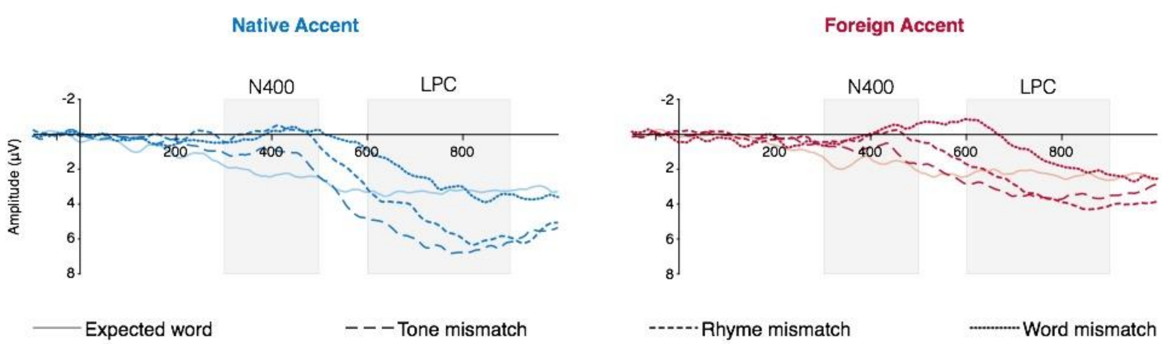

Figure 5. Grand average waveforms averaged across posterior electrodes (low-pass filter of 20Hz applied for visualization only), highlighting the N400 (300-500ms) and LPC (600-900ms) windows (waveforms for individual electrodes available in Appendix S2)

Table 3. Mean amplitude and standard error for the N400 window (300-500 ms)

	expected word	tone mismatch	rhyme mismatch	word mismatch
native accent	2.22 (0.11)	1.18 (0.11)	-0.23 (0.11)	-0.12 (0.10)
foreign accent	1.58 (0.11)	1.11 (0.11)	0.06 (0.11)	0.06 (0.10)

Mean response (in µV); standard errors indicated in parentheses.

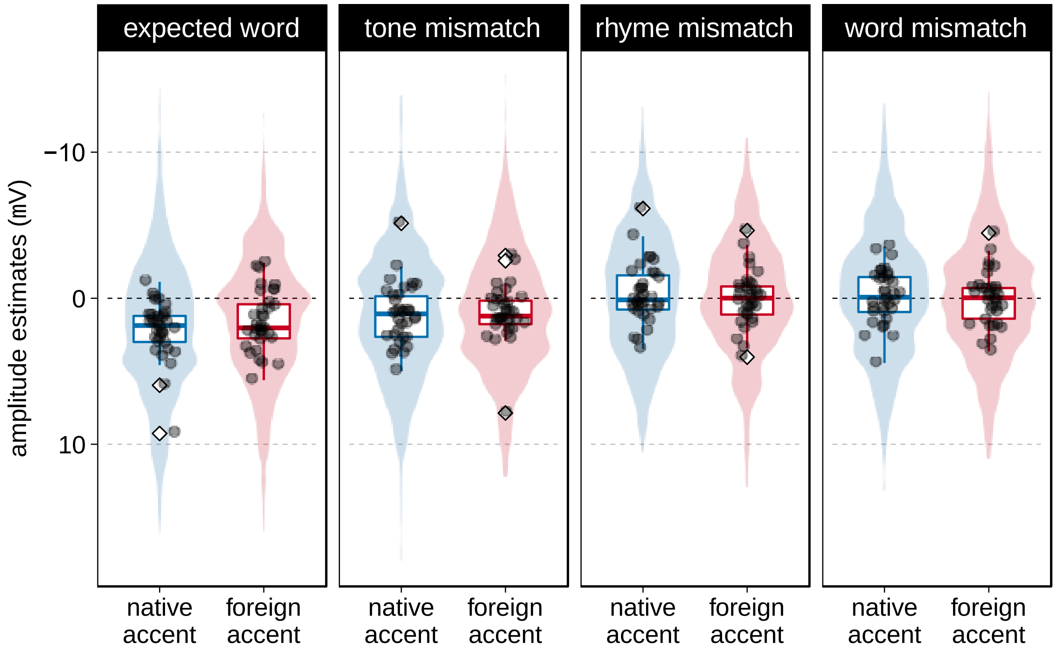

Figure 6. Boxplots depicting the model estimates (amplitude) for the N400 (300-500ms) window over posterior electrodes. Shaded areas behind the boxplots indicate the distribution of estimates. Grey dots indicate the mean amplitude of each participant's responses; white diamonds indicate that a participant's mean is an outlier.

To answer our research questions, we conducted post-hoc comparisons, reported in Table 4. In all comparisons, *b* values can be interpreted as amplitude differences (in µV). In order to capture overall trends, especially with respect to N400 effects, we tested effects of the mismatch conditions across accents. There were statistically significant N400 effects in response to rhyme and word mismatches, with amplitudes more negative than those of expected word responses. Relative to other conditions, the 95% confidence intervals (CIs) are narrow, with upper bounds falling approximately 1.7 µV below zero. The difference between responses to expected words and tone mismatches failed to reach statistical significance, while the difference between rhyme and tone mismatches was statistically significant. In other words, we found relatively strong N400 effects (>3.5 µV) for word and rhyme mismatches, but failed to find N400 effects for tone mismatches, regardless of accent.

Table 4. Post-hoc comparisons for N400 model

	b	*SE*	*z*	*Pr(>|z|)*	Conf. Intervals	
					Lower	Upper
Across accents						
Tone mismatch – Expected word	-1.58	0.76	-2.09	.185	-3.66	0.49
Rhyme mismatch – Expected word	-3.83	0.78	-4.90	<.001	-5.96	-1.69
Word mismatch – Expected word	-3.79	0.77	-4.90	<.001	-5.90	-1.67
Rhyme mismatch – Tone mismatch	-2.25	0,76	-2.95	.019	-4.33	-0.17
Interactions of accent effects						
(Nat: Tone-Exp) – (For: Tone-Exp)	-0.25	0.90	-0.27	1.000	-2.71	2.22
(Nat: Rhyme-Exp) – (For: Rhyme-Exp)	-0.68	0.87	-0.79	1.000	-3.05	1.69
(Nat: Word-Exp) – (For: Word-Exp)	-0.61	0.87	-0.70	1.000	-2.99	1.76
(Nat: Tone-Rhyme) – (For: Tone-Rhyme)	0.44	0.79	0.55	1.000	-1.74	2.61

p-values adjusted by Bonferroni-Holm method; asymptotic confidence intervals reported
Nat = Native accent; For = Foreign accent;
Exp = expected word; Tone = tone mismatch; Rhyme = rhyme mismatch; Word = word mismatch

Post-hoc comparisons of the interactions of accent effects indicate no significant interaction effects. So, with respect to our first question about responses to foreign-accented pronunciation errors, we failed to find evidence of differences between accent conditions. If this null result reflects a true lack of meaningful differences between accent conditions, this would suggest foreign accent does not strongly impact N400 responses. However, results should be treated as very uncertain, as wide CIs suggest a large degree of imprecision.

For our second research question, we did find evidence of differences in responses to tone and rhyme mismatches—but no accent effect. In other words, regardless of accent, listeners showed N400 effects for rhyme errors, but no—or at least very small—responses to tone errors. Again, in light of the wide CIs, results should be treated as uncertain.

LPC

Mean amplitudes for word and accent conditions in the LPC window are shown in Table 5. Mean results suggest more positive amplitudes for native accented speech across critical word conditions, and more positive amplitudes for tone and rhyme mismatches within both accent conditions. Across all conditions, the amplitude of LPCs for foreign-accented speech is about 1-1.5 µV less positive than those for native speech. A mixed model ANOVA showed significant main effects for word, $F(3, 96.85)=17.34$, $p<.001$, and accent, $F(1, 109.69)=30.77$, $p<.001$. The word-by-accent interaction was not significant, $F(3, 74.17)=1.57$, $p=.203$. Model estimates are depicted in Figure 7.

Table 5. Mean amplitude and standard error for the LPC window (600-900 ms)

	expected word	tone mismatch	rhyme mismatch	word mismatch
native accent	2.96 (0.13)	5.77 (0.13)	4.51 (0.13)	2.44 (0.12)
foreign accent	2.05 (0.12)	3.43 (0.12)	3.00 (0.12)	1.10 (0.12)

Mean response (in µV); standard deviations indicated in parentheses.

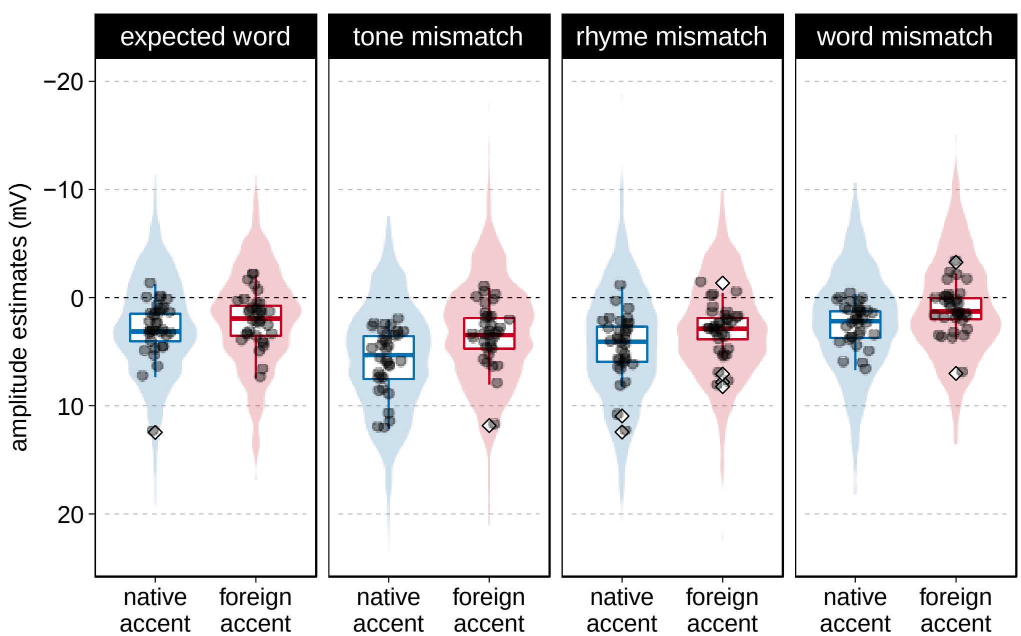

Figure 7. Boxplots depicting the model estimates (amplitude) for the LPC (600-900ms) window over posterior electrodes. Shaded areas behind the boxplots indicate the distribution of estimates. Grey dots indicate the mean amplitude of a single participant's responses; white diamonds indicate that a participant's mean is an outlier.

Post-hoc comparisons are reported in Table 6. In order to capture overall LPC trends, we tested effects of the mismatch conditions across accents. There were statistically significant LPC effects in response to tone and rhyme mismatches, with amplitudes more positive than those of expected word responses. Wide CIs suggest a large degree of uncertainty, with plausible values (under the given model's assumptions) ranging from fairly weak effects (<1 μV) to very large effects (>6.5 μV). The difference between responses to expected words and word mismatches failed to reach statistical significance, and there was also no statistically significant difference between tone and rhyme mismatch LPCs.

Table 6. Post-hoc comparisons for LPC model

					Conf. Intervals	
	b	*SE*	*z*	*Pr(>|z|)*	Lower	Upper
Across accent						
Tone mismatch – Expected word	4.15	0.93	4.46	<.001	1.60	6.69
Rhyme mismatch – Expected word	2.62	0.85	3.09	.014	0.30	4.94
Word mismatch – Expected word	-1.54	0.79	-1.95	.259	-3.71	0.63
Rhyme mismatch – Tone mismatch	-1.52	0.97	-1.57	.469	-4.19	1.14
Interaction of accent effects						
(Nat: Tone-Exp) – (For: Tone-Exp)	1.83	0.87	2.10	.214	-0.55	4.22
(Nat: Rhyme-Exp) – (For: Rhyme-Exp)	0.79	0.94	0.85	.845	-1.77	3.35
(Nat: Word-Exp) – (For: Word-Exp)	0.67	0.96	0.70	.845	-1.95	3.29
(Nat: Tone-Rhyme) – (For: Tone-Rhyme)	1.04	0.97	-1.08	.845	-3.68	1.60

p-values adjusted by Bonferroni-Holm method; asymptotic confidence intervals reported
Nat = Native accent; For = Foreign accent;
Exp = expected word; Tone = tone mismatch; Rhyme = rhyme mismatch; Word = word mismatch

Though there was a main effect of accent on LPC responses, post-hoc comparisons of the interactions of accent effects indicated no significant differences. So then, with respect to our first research question, we failed to find evidence of an accent effect on listener responses to pronunciation errors. As with N400 results, wide CIs suggest a large degree of imprecision and uncertainty.

Post-hoc comparisons also targeted the critical comparison of LPC responses to tone and rhyme mismatches. We found no significant differences either across or between accents, though descriptively, tone mismatch responses were more strongly positive than rhyme mismatch responses, and the magnitude of both mismatch responses was larger for native accented speech.

General Discussion

We conducted an ERP study with native listeners of Chinese, who judged the accuracy of sentences produced by either a native speaker of Mandarin, or L2 Mandarin speaker with a foreign (American English) accent, and we compared N400 and LPC responses to critical words that were expected, mispronounced by either a tone or a rhyme (thus becoming a nonword), or were contextually inappropriate. We wanted to know whether listeners' expectations for more pronunciation errors in foreign-accented speech would result in reduced ERP responses to such errors (relative to native-accented speech), and whether this effect would be larger for tone than rhyme errors.

To summarize our findings, (1) behavioral judgments of sentences indicated less accuracy for tone mismatches when produced by the foreign-accented speaker, and less accuracy to tone mismatches (relative to rhyme mismatches) regardless of accent. We also found a large accent effect for the expected word condition, with listeners less likely to accept sentences produced by the L2 speaker. (2) There were statistically significant N400 effects for rhyme and word mismatches, but not for tone mismatches; there was also a significant difference between tone and rhyme mismatch N400s. There were no significant interactions with accent conditions. (3) LPCs for foreign-accented speech had less positive deflections overall, and there were significant LPCs for tone and rhyme mismatches in both accent conditions, but there were no significant interactions with accent conditions. (4) Finally, both behavioral and ERP data suggest a high degree of variability in listener responses. Below, we will discuss these results in more detail, tying together behavioral and ERP results.

The role of foreign accent in Mandarin speech processing

Our results provide some evidence that foreign-accent may shape how Mandarin listeners process pronunciation errors. Behavioral judgments suggest that many listeners were more accepting of sentences with critical tone mispronunciations in foreign-accented speech than in native speech. What might explain this difference?

One possible explanation is that upon identifying a speaker as non-native, listeners used their knowledge of the world to adapt their expectations. Knowing that tone errors are typical in foreign-accented Mandarin, listeners might choose to apply less stringent criteria for 'error' when judging an L2 speaker's tones.

A second possible explanation is that, after hearing enough tone errors, listeners might begin to change the way they utilize acoustic-phonetic cues (i.e., F0 or other features of tones) in the speech of a foreign-accented speaker. Once they noticed that tones were often unhelpful or even misleading when trying to recognize words, the listener may have begun to rely on tones less, and use other available cues (e.g., segments, context) more. Idemaru & Holt (2011) showed that listeners will down-weight some types of redundant acoustic-phonetic features if those features stop being informative. This down-weighting might be a way to 'ignore' pronunciation errors, however, it is unclear if it can occur for *phonological* features (Pelzl et al., 2020). Many studies have shown listener adaptation to accented features of L2 speech (e.g., Baese-Berk et al., 2013; Bradlow & Bent, 2008; Reinisch & Holt, 2014; Xie et al., 2017), but these accented pronunciation features were still *informative*. That is, by adapting, listeners become more efficient at comprehending L2 speech. In the case of pronunciation errors, however, it will usually be better to ignore the error entirely, rather than trying to interpret it. Thus, it may be that listeners accepted more tone mismatch sentences from the L2 speaker because they began to down-weight his tones.

Finally, a third explanation for behavioral results is that phonetic information in foreign-accented speech is more difficult for listeners to process, leading to decreased accuracy in judgment of pronunciation errors. Relative to rhyme errors (or at least those in our stimuli), tone errors might be more subtle, leading to even lower accuracy in judgments of tones compared to rhymes. This trend has been observed previously with native speech (Cutler & Chen, 1997; Pelzl et al., 2019), and in the present study is supported by lower rejection of tone mismatches for *both* speakers. Perhaps, then, foreign-accent simply makes detection of tone errors even more difficult.

As noted above, we recommend caution in interpretation of the present sentence judgment results, given that responses cannot be unambiguously tied to the mismatch manipulations. Future work should use less ambiguous measures to establish whether the behavioral patterns observed here are reliable.

If behavioral results suggest effects for foreign-accented speech, ERP results are less clear. N400s showed no evidence of an effect for accent. LPCs showed a main effect of accent, but no interactions with mismatch conditions. In other words, for LPCs there appears to be a difference of degree in response to foreign-accented speech errors compared to native speech errors, regardless of the type of error. Superficially this might look inconsistent with previous studies which found significantly reduced LPCs to foreign-accented speech errors (Caffarra &

Martin, 2018; Hanulíková et al., 2012; Romero-Rivas et al., 2015), but despite a failure to reach statistical significance, our trends are clearly in the same direction as those studies, suggesting somewhat attenuated LPCs for foreign-accented speech—both for tone and rhyme mismatches.

Recent work on LPC/P600 effects has highlighted that, in many cases, these late responses are largely indistinguishable from the P3b response (e.g., Leckey & Federmeier, 2019). Importantly, the amplitude of the P3b is sensitive to salience, such that stimuli that are more salient elicit larger P3bs. In this case, weaker responses for foreign-accented pronunciation errors might indicate *less salience* for these deviations, relative to native speech. This line of interpretation accords well with the reasoning presented by Hanulíková and colleagues (2012), who argued that a lack of N400 differences for lexical-semantic violations indicated that listeners had little difficulty comprehending the speaker's message; instead, it was largely their different expectations for usage of grammatical gender that reduced LPCs for foreign-accented speech. These ideas also fit well with models of speech perception such as the ideal adapter (Kleinschmidt & Jaeger, 2015) or noisy channel (Gibson, Bergen, & Piantadosi, 2013; Gibson et al., 2017), which make explicit the role of listeners' high-level expectations in driving responses.

While our data comport with this line of thinking, we cannot rule out alternative accounts that would attribute trends in behavioral and neural responses to difficulties perceiving the accented speech itself. In the present study, it could also be the case that listeners in fact detected more frequent pronunciation errors in foreign-accented speech than in native speech—that is, they may have detected errors *in addition* to the critical mismatchting words, and done so even in the expected word sentences. Importantly, increasing the frequency of task-relevant events is known to attenuate the P3 (Leckey & Federmeier, 2019; Squires, Wickens, Squires, & Donchin, 1976). Thus, unintended pronunciation deviations by the foreign-accented speaker may also have impacted LPC responses by increasing the overall frequency of 'errors' in the foreign-accented stimuli. Finally, there is no reason pronunciation errors would have to function in the same way that syntactic violations have in previous studies.

Future work along these lines should seek to more effectively tease apart alternative accounts, exercising more control on the properties of foreign-accented speech. The use of judgment tasks or their avoidance should also be carefully considered, as such tasks no doubt impact ERP responses, even when decisions are delayed, as they were in the current study. If we believe the LPCs we are interested in are, in fact, P3b responses, then the wealth of relevant literature can help guide design choices.

Potential differences between rhymes and tone mismatches

In addition to the accent-specific behavioral effect for tone mismatches, we also found a more general behavioral effect for tone mismatches. That is, regardless of a speaker's accent, when attempting to judge sentences for pronunciation deviations, listeners were more consistent at correctly rejecting rhyme than tone mismatches. These behavioral differences were mirrored by N400 effects in ERPs. In both accent conditions, tone N400 effects were weaker than rhyme N400 effects, and failed to differ from expected word N400s. Though we did not set out to test this difference in native speech, the finding is somewhat novel, and requires a bit of discussion.

First, our results are consistent with behavioral outcomes from Pelzl et al. (2019), who found less accuracy for tone mismatches relative to rhymes both in a disyllabic lexical decision task, and a sentence judgment task that used a subset of the stimuli used here (with only the native Mandarin speaker). However, consistent with other ERP studies reviewed above (Brown-

Schmidt & Canseco-Gonzalez, 2004; Schirmer et al., 2005), Pelzl et al. (2019) failed to find statistically significant ERP differences for tone and rhyme mismatches.

Given the close relationship between this study and that of Pelzl et al. (2019), the difference in ERP results is unexpected. There are three main differences between studies. First, errors were overall more frequent in this study (75% overall) compared to in Pelzl et al. (2019: 50% overall). This did not seem to impact behavioral judgments, as Pelzl et al. (2019) reported 16% acceptance rates for tonal mismatches and 5% for rhymes, compared to 14% tone and 7% rhyme acceptance for native-accented mismatches in the present study. Second, the number of stimuli in the present study (240 sentences) was greater than that in Pelzl et al. (2019: 180 sentences). This meant we had a more relaxed criterion for cloze probabilities of critical words in the present study. This difference should have been addressed by counter-balancing across lists, so that any affects would be similar for tone and rhyme mismatch conditions. Finally, the present study had both a native and a foreign-accented speaker. Perhaps encountering the accented speaker in the stimuli affected the expectation for tone errors globally—and may have in fact increased it if there were incidental tone errors in sentences. This could have reduced sensitivity for all tone mismatches, regardless of accent. Clearly, these results will need to be replicated, and further work will be needed to determine the cause or causes of N400 differences.

Individual differences

A clear feature of the present data is large variability among listeners, especially for the foreign-accented speaker in the expected word condition of the behavioral task. Figure 8 illustrates visible trends in this variability, showing that about a third of participants (solid black lines) tended to reject most sentence spoken by the foreign-accented speaker, even if the critical

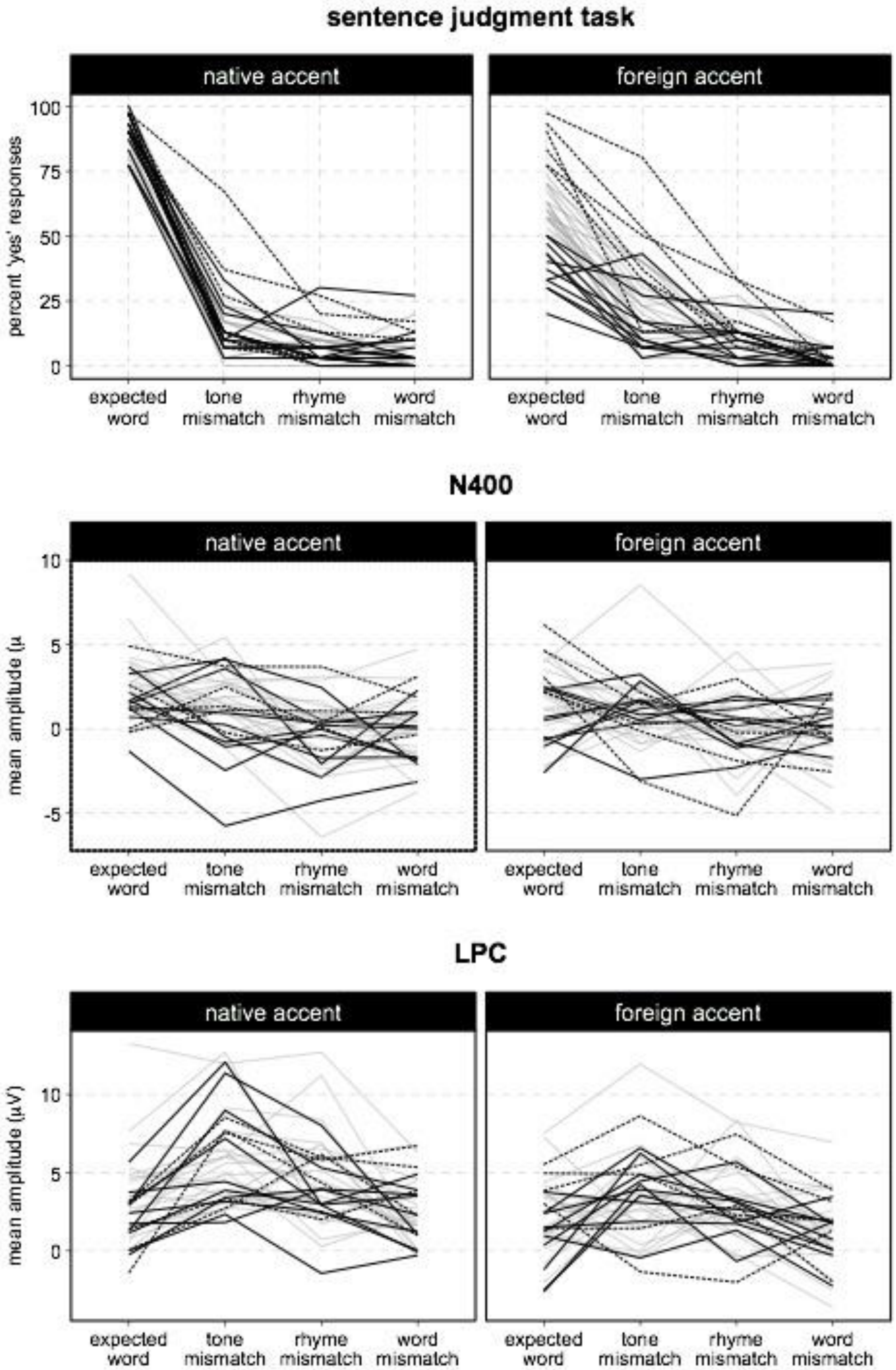

Figure 8. Individual participant patterns across measures. Each line indicates a single pariticipant's mean (percent or amplitude) across critical word conditions. Line type is associated with performance in the expected word sentences for the foreign-accented speaker. Solid black lines indicate <50% 'yes' responses Dashed black lines indicate >75% 'yes' responses, Grey lines indicate more moderate judgments.

word was the expected word. On the other extreme there were a handful of participants (dashed lines) who appeared to accept almost every expected word and most tone mismatch sentences from the foreign speaker. Patterns in the ERP data are not so distinctive—though variability is still clearly a feature of these data as well.

Previous ERP studies (Caffarra & Martin, 2018; Grey & van Hell, 2017) found that people's (lack of) experience with foreign-accented speakers was connected to their ERP responses to foreign-accented speech errors. We might then speculate that individual participant differences in our behavioral and ERP data relate to their previous experience with L2 Mandarin speakers (i.e., listeners with more experience and thus stronger expectations for L2 tone errors might have reduced N400/LPC responses to those errors). Unfortunately, we did not gather this type of background information from our participants.

In addition to participants' experience with foreign-accented Mandarin speakers, their experience with native, but *regionally-accented* Mandarin speakers could also have led to individual differences in task performance. China's native language environment is highly diverse, with at least seven major Chinese language varieties (e.g, Mandarin, Cantonese, and Shanghainese), and almost countless local dialects of each (Norman, 1988). While Mandarin is the common language, people's other Chinese languages can have strong influences on their Mandarin pronuncation. Individual differences in experience with these many flavors of Mandarin might influence sensitivity to tone (and rhyme) deviations from the 'standard' pronunciation. Regional differences of this sort have been previously shown to impact outcomes in lexical tasks (Wiener & Ito, 2016). Future work might explore these issues further, perhaps contrasting listeners' processing and judgments of regional and foreign-accented Mandarin speech.

Finally, we cannot rule out a strategic component to participants' behaviors. Even though they were told specifically what types of speech errors should lead to rejections, some participants may have settled on differing strategies to carry out the judgment task. For example, some may have focused more heavily on individual words, while others tried to take in the sentences more holistically. This could certainly have influenced outcomes, especially at the behavioral level.

Although we cannot provide a satisfactory account of the variability in our data, such variability is not necessarily surprising. Work by Tanner & Van Hell (2014) and Tanner (2019) illustrates how simple measures of central tendencies at the group level can often misrepresent individual ERP patterns. We also acknowledge that variability is sometimes an indication of noisy data. Given these realities, the presentation of estimates of variability (SE, CI) should be a standard feature of ERP studies, in order to help researchers determine how closely results reflect reliable group trends, and to allow for comparison of variability across studies.

Limitations and future directions

We would be remiss not to point out some limitations of the present work, as these may be key areas for improvement in future work.

First, this study used just a single L2 Mandarin speaker. It is possible this speaker's Mandarin accent is not representative. A small post-hoc accentedness rating task suggests he had a moderate to strong accent (see Appendix S1 for details). Future work should follow the example of other studies by establishing accentedness a priori (cf. Grey & van Hell, 2017), and perhaps utilizing a variety of L2 speakers (Romero-Rivas et al., 2015).

Although the sample size of the current study compares well with other accented speech and Chinese tone ERP studies, this does not mean it was sufficient to find relevant effects. Future studies should strive to substantially increase participant numbers (Brysbaert & Stevens, 2018), and also to reduce the complexity experimental designs when possible (Luck & Gaspelin, 2017).

Our participants (young, urban, college educated) cannot be considered representative of the Chinese population more broadly. Future work might make efforts to expand the population of Chinese people recruited to join research.

Finally, as noted in our review, Pelzl et al. (2019) found visible differences between word and nonword N400 effects, with word mismatch N400s continuing much longer than those of nonwords. These trends were replicated here, suggesting there may be distinctive N400 effects for words compared to nonwords. The current study was not designed to pinpoint the reasons for such differences. The extended N400 for word mismatches, compared to both tone and rhyme nonwords, may be due to the disambiguating information provided by the following syllable, which could allow recovery of the originally expected lexical target. Alternatively, the quickly attenuating N400 response for nonwords might indicate a failure to access a word at all. Finally, it could be that the key difference is indexed not in the N400, but in the strong LPC that follows for nonwords. Future research might attempt to distinguish between pronunciation errors that lead to words compared to nonwords, while attempting to control possible confounds such as word length (monosyllable vs disyllable). By taking advantage of the distinctive properties of monosyllabic words (few nonword neighbors) and disyllabic words (many nonword neighbors), we may find ways to disentangle effects that are otherwise confounded. Pursuing this line of work in the context of foreign-accented speech processing might be optimal as speech errors will be ecologically valid, and could provide a more realistic assessment of the importance of accurate tones (or segments) than is typically accomplished by artificial manipulations of native speech.

Conclusion

The present study set out to explore the relationship between second language accent and pronunciation errors in the context of Mandarin Chinese sentence comprehension. Results indicate potential differences in the way Mandarin listeners judge and process foreign-accented compared to native-accented pronunciation errors. They also suggest tone mismatches may be processed differently from rhyme mismatches, regardless of accent. Finally, we found evidence of variability across listeners. Such variability may be a valuable object of study in future work, as it could provide additional insight into the factors that influence foreign-accented speech processing in Mandarin.

Notes

[1] There was one extreme outlier who accepted 80% of tone mismatches in the foreign accent, and 67% in the native accent. To be certain results were not due soley to this participant, we removed them and reran the analysis. There was no substantive difference in outcomes, except that the interaction of accent effects between rhyme and word was not significant (b = -1.39; SE = .592; z =-2.36; p=. 074).

[2] An additional analysis using anterior electrodes is reported in the Appendix S4. An earlier attempt to model electrodes nested under subjects was abandoned due to model fitting difficulties.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Additional details for stimuli design and creation

Appendix S2: Waveforms for all critical electrodes

Appendix S3: Full mixed-effects model output

Appendix S4: Additional analyses

Appendix S5: List of stimuli